

All questions are single-choice. Each question is worth 5 points, for a total of 20 questions.

1. A particle of mass m slides down a smooth surface described by $y = ax^2$ from a height H . What is the magnitude of the normal force at the lowest point?
(A) mg (B) $mg + 2aHm$ (C) $mg + mv^2/R$ (D) mv^2/a (E) $(mg^2 + 4aHmg)^{1/2}$
2. A simple spring-mass system consists of a block of mass m attached to a spring with spring constant k . What is the speed of the block as it passes through the equilibrium position x_0 , given that the initial displacement is A ?
(A) $A(kx_0^2/m)^{1/2}$ (B) $A(k/m)^{1/2}$ (C) $1/2(kx_0^2)$ (D) $(2gA)^{1/2}$ (E) kx_0
3. For the potential energy function $V(x) = -kx + \lambda x^3$, what is the nature of the equilibrium point(s)?
(A) Stable (B) Unstable (C) Neutral (D) No equilibrium (E) Two stable and one unstable point
4. A satellite of mass m is mounted on a rocket that is initially at rest on the surface of the Earth. If the Earth has mass M , what is the total energy the rocket must provide in order for the satellite to orbit at an altitude equal to R_e above the Earth's surface?
(A) GMm/R_e (B) $1/2(GMm/R_e)$ (C) $3/4(GMm/R_e)$ (D) $1/2(GMm/R_e^2)$ (E) R_e/GMm
5. A disk with a moment of inertia I and initial angular velocity ω_0 is slowed down by a constant torque τ . What is the stopping time?
(A) ω_0/τ (B) $I\omega_0/\tau$ (C) τ/ω_0 (D) $\omega_0^2/2\tau$ (E) $\omega_0 I / \tau^2$
6. An ideal gas undergoes a closed isobaric cycle with a triangular P - V diagram. If the base is ΔV and the height is P , what is the net work done?
(A) $P\Delta V$ (B) $P\Delta V/2$ (C) $3P\Delta V/2$ (D) 0 (E) $2P\Delta V$
7. The electric field is given by $E(x) = E_0 e^{-x/\lambda}$. What is the potential difference from $x = 0$ to ∞ ?
(A) ∞ (B) $E_0\lambda$ (C) $-E_0\lambda$ (D) 0 (E) $\lambda^2 E_0$
8. Four point charges of $+0.500 \mu\text{C}$, $+1.50 \mu\text{C}$, $-1.00 \mu\text{C}$, and $-0.500 \mu\text{C}$ are randomly distributed on a ring. If the electric potential at the center of the ring due to the $+0.500 \mu\text{C}$ charge alone is $4.5 \times 10^4 \text{ V}$, what is the total electric potential at the center due to all four charges?
(A) 0 (B) $-4.5 \times 10^4 \text{ V}$ (C) $4.5 \times 10^4 \text{ V}$ (D) $9.0 \times 10^4 \text{ V}$ (E) $18.0 \times 10^4 \text{ V}$
9. A charged particle with mass m and charge q enters a magnetic field B at angle θ with velocity v . What is the radius of the helical trajectory?
(A) $mv/(qB)$ (B) $mv\sin\theta/(qB)$ (C) $mv\cos\theta/(qB)$ (D) $qB/(mv)$ (E) $\lambda/(2\pi)$

注意：背面有試題

10. An infinitely large, thin conducting sheet carries a uniform current density J . What is the magnetic field at a distance r perpendicular to the surface of the sheet?
(A) $1/2 \mu_0 J$ (B) $1/2 \mu_0 J r$ (C) $\mu_0 J$ (D) $\mu_0 J r$ (E) $1/2 \mu_0 J^2$
11. The magnetic flux varies with time as $\Phi(t) = \Phi_0 e^{-bt}$. What is the maximum induced emf?
(A) $b\Phi_0$ (B) Φ_0/b (C) 0 (D) $\Phi_0^2 b$ (E) Φ_0/b^2
12. What is the relationship between the power P emitted by a blackbody and its temperature T ?
(A) $P \propto T$ (B) $P \propto T^2$ (C) $P \propto T^3$ (D) $P \propto T^4$ (E) $P \propto T^5$
13. In the photoelectric effect, which equation relates stopping voltage U_0 and wavelength λ ?
(A) $hc/(e\lambda) = U_0$ (B) $eU_0 = hc/\lambda$ (C) $U_0 = hc/(e\lambda) - \phi$ (D) $U_0 = \phi - hc/(e\lambda)$ (E) $\phi = U_0$
14. In a hydrogen atom, what is the energy difference between $n = 3$ and $n = 2$?
(A) 13.6 eV (B) 1.89 eV (C) 3.4 eV (D) 10.2 eV (E) 12.1 eV
15. An observer measures a clock moving at speed $v = 0.6c$. What is the time dilation factor γ ?
(A) 1.0 (B) 1.15 (C) 1.25 (D) 1.50 (E) 2.0
16. What is the energy density ratio of electric and magnetic fields in an electromagnetic wave?
(A) Equal (B) Electric is twice magnetic (C) Magnetic is twice electric (D) Cannot be compared (E) Depends on frequency
17. A radio station requires a coaxial cable to connect the transmitter to the antenna. The cable consists of an inner conductor with a diameter of 0.2 mm and an outer conductor with a diameter of 2.0 mm, whose thickness is negligible. If the maximum safe potential difference between the conductors is 2.3 kV, what is the maximum charge density the conductors can carry under this safety constraint?
(A) $45 \mu\text{C}$ (B) $50 \mu\text{C}$ (C) $55 \mu\text{C}$ (D) $60 \mu\text{C}$ (E) $65 \mu\text{C}$
18. A spreading wave packet increases in width over time due to which principle?
(A) Second Law of Thermodynamics (B) Conservation of Energy (C) Relativistic Effect (D) Uncertainty Principle (E) Momentum Conservation
19. A particle moves in a non-conservative field but its total energy remains conserved. What is the likely form of the external force?
(A) Constant force (B) Position-dependent (C) Velocity-dependent (D) Perpendicular to displacement (E) Time-dependent
20. When a photon undergoes Compton scattering with an electron and is scattered at a 90° angle relative to its original direction, its energy after scattering is half of its original energy. What is the initial wavelength of the photon?
(A) 1.33 nm (B) 2.43 nm (C) 1.33 pm (D) 2.43 pm (E) None of the above